



Chemical Theory II

Statistical

Thermodynamics

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The course is developed based on the
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Summary of last lecture

- Vibrational partition function
 - Harmonic oscillator: $E_v = (v + 1/2) h\nu_0$
 - Partition function is a geometric progression.
 - Vibrational temperature: $\theta_{vib} = h\nu_0 / k$
 - Often $q_{vib}' \approx 1$
 - Normal modes for polyatomics
- Electronic partition function
 - Often dominated by the ground state.
 - Degeneracy of $(2J + 1)$ for atoms
- Separate contributions to thermodynamic functions
 - Associate the $N!$ term with translational partition function.

Course outline

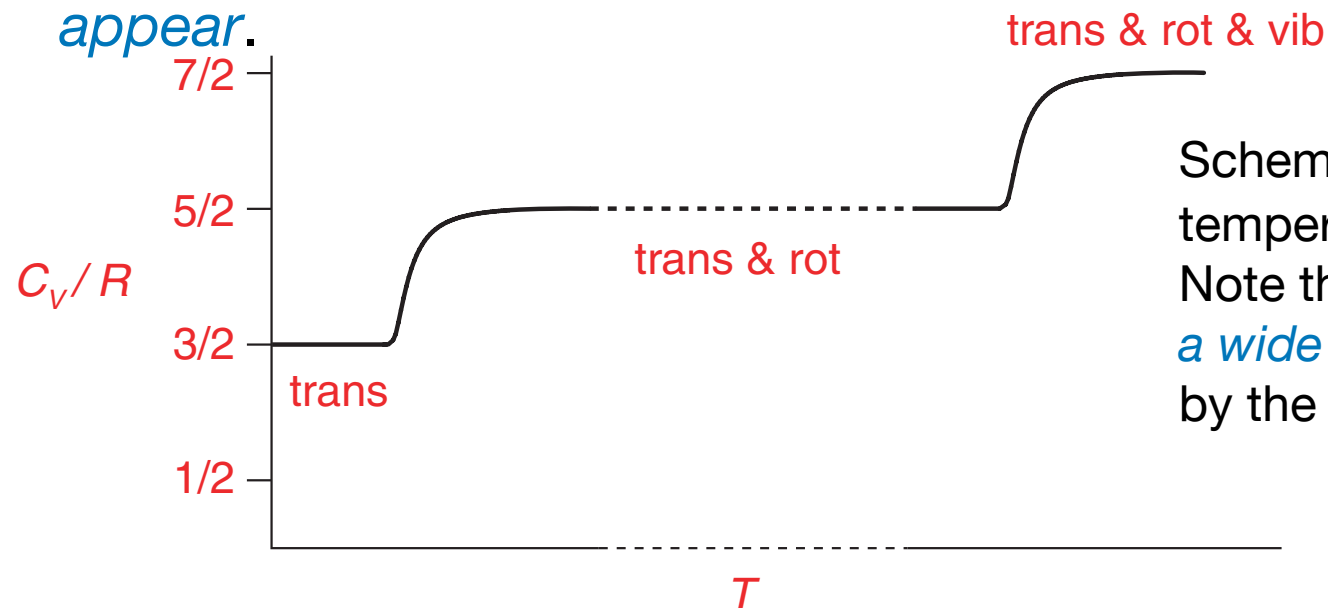
- Internal energy and heat capacities

Internal energy and heat capacities

- In this section we will look at the contributions that different kinds of energy levels make to the internal energy, U , and the (constant volume) heat capacity, C_V . This will be the first practical application of the relationships derived in the preceding lectures, and it will also give important insight into the relationship between partition functions and thermodynamic variables.

Internal energy and heat capacities

- One of the observations which we will be able to explain is the rather characteristic way in which *heat capacities of gases vary with temperature*. Typically, what is found is that *as the temperature is increased the heat capacity changes little until a particular point is reached at which the heat capacity quickly rises to a new value; after this it once again remains constant as the temperature increases*. At still higher temperatures a further step occurs. The result is *a series of plateaus*, illustrated below for a diatomic molecule. What is more remarkable is that *the plateau values are the same for all diatomic molecules; all that changes is the temperatures at which the steps appear*.



Schematic variation of C_v with temperature for a diatomic molecule. Note that *the central plateau covers a wide temperature range*, indicated by the dashed line.

Internal energy and heat capacities

- The interpretation of this behaviour is that *at the lowest temperatures only translational energy levels are occupied significantly and so only these contribute to the heat capacity. Eventually the temperature becomes high enough for rotational levels to be occupied and as a result the heat capacity increases. Finally, at sufficiently high temperatures vibrational energy levels are occupied and the heat capacity rises once more.*
- By computing the heat capacity in terms of the partition function we will be able to predict the form of this curve, explain why there are a series of plateaus, find out the value of the heat capacity at the plateaus and identify the temperatures at which the steps occur. This discussion will lead us to an understanding of the *equipartition principle* and how it can be applied.

Equipartition principle

- The *equipartition principle* states that *each ‘squared term’ contributes $1/2 RT$ (per mole) to the internal energy of the system*; **R** is the gas constant, in SI given in units of $\text{J K}^{-1} \text{mol}^{-1}$. In molecular units the contribution is *$1/2 kT$ per particle*.
- What is a ‘*squared term*’? *It is any term in the expression for the energy of the molecule or atom which depends on the square of a position or velocity variable.*
- The translational energy depends on the square of the velocity, and so qualifies as a squared term. *Motion in the x , y , and z directions is independent, so translation in three-dimensions gives three squared terms and thus an internal energy of $3/2 RT$. This is often expressed by saying that there are three translational degrees of freedom.*

Equipartition principle

- The rotational energy depends on the square of the angular velocity and so qualifies as a squared term. In general a molecule can rotate about three orthogonal axes; each rotation gives rise to a contribution of $1/2 RT$. *There are thus three rotational degrees of freedom, contributing a total of $3/2 RT$. A diatomic (or any linear molecule) is a special case as there is no rotation about the internuclear axis, so there are only two rotational degrees of freedom. The internal energy is thus RT .*
- A vibration has two squared terms: one for the kinetic energy, which goes as the square of the velocity, and one for the potential energy, which goes as the square of the displacement. *The vibration of a diatomic molecule thus contributes RT to the energy. In general, each normal mode contributes RT to the energy.*

Heat capacities

- If the translational contribution to U is $\frac{3}{2} RT$, the corresponding contribution to the heat capacity is

$$C_V = \left(\frac{\partial U}{\partial T} \right)_V = \frac{d \frac{3}{2} RT}{dT} = \frac{3}{2} R.$$

Likewise, *for a diatomic, the rotational contribution is R and the vibrational contribution is also R .*

- We can now explain the form of the plot on the previous page. At the lowest temperatures only *translational levels* contribute and so the heat capacity is $\frac{3}{2} R$; this rises to $\frac{5}{2} R$ once the temperature is high enough for *the rotational levels* to be occupied and so contributing a further R to the heat capacity. At still higher temperatures, *vibrational levels* begin to be occupied, ultimately contributing a further R to the heat capacity and bringing the final value to $\frac{7}{2} R$.

Example

- Exercise: show that for H₂O $C_v = 3R$ if the vibrations do not contribute, and $= ?$ if they do.

Translational contribution

- The partition function for translation in three dimensions is

$$q_{\text{trans}} = \left(\frac{2\pi mkT}{h^2} \right)^{\frac{3}{2}} V.$$

- Looking at the previous results we can see that to find U we need the derivative of q with respect to temperature at constant volume. The only term which depends on T under these circumstances is $T^{3/2}$, so q_{trans} can be written

$$q_{\text{trans}} = \alpha T^{\frac{3}{2}}$$

where α is a constant.

Translational contribution

- U_{trans} is then easy to calculate as

$$\begin{aligned}U_{trans} &= NkT^2 \left(\frac{\partial \ln q_{trans}}{\partial T} \right)_V \\&= NkT^2 \left(\frac{\partial \ln \alpha T^{\frac{3}{2}}}{\partial T} \right)_V = NkT^2 \left(\frac{\partial (\ln \alpha + \frac{3}{2} \ln T)}{\partial T} \right)_V \\&= \frac{3}{2} NkT^2 \frac{1}{T} = \frac{3}{2} NkT\end{aligned}$$

where we have used the property that α is independent of T at constant V . For one mole of particles, U_{trans} is $\frac{3}{2} N_A kT = \frac{3}{2} RT$ and so $C_{V,trans} = \frac{3}{2} R$; these are precisely the values predicted by the equipartition principle.

- The reason that the value of U is in accord with that predicted by the equipartition principle is that *the expression for q_{trans} was derived under the assumption that the energy level spacing was much less than kT . It is precisely under these circumstances that the equipartition values of the internal energy apply.*

Rotational contribution

- Previously we found that when the rotational energy level spacing was small compared to kT , the rotational partition function for a diatomic molecule is given by

$$q_{\text{rot}} = \frac{T}{\sigma \theta_{\text{rot}}}$$

- As with translation, this takes the form αT , where α is *a temperature-independent constant*. The rotational contribution to the internal energy is thus

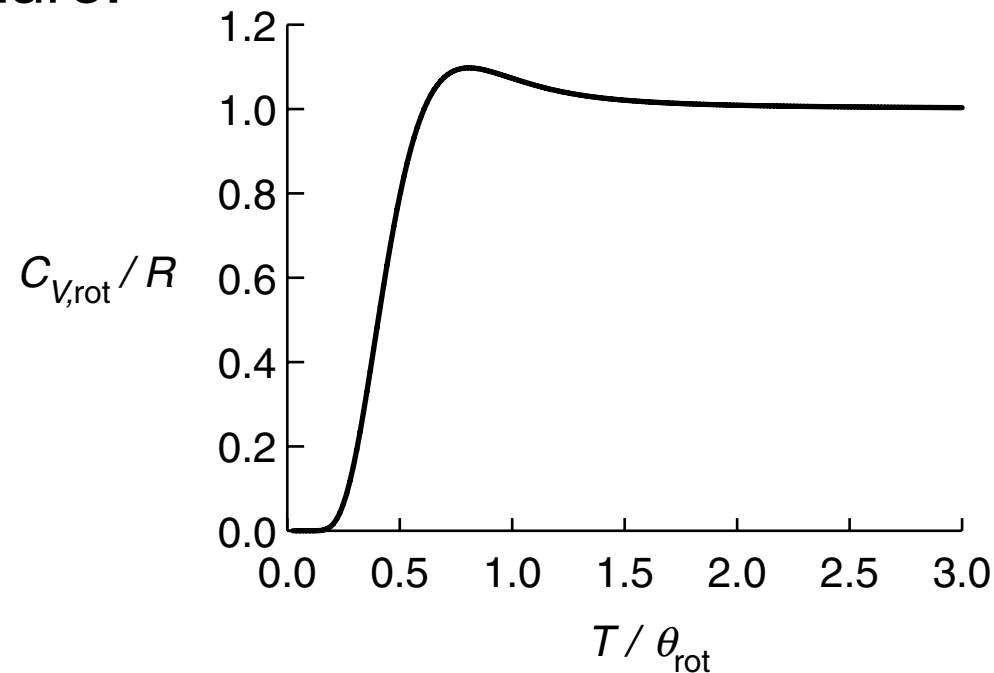
$$\begin{aligned} U_{\text{rot}} &= NkT^2 \left(\frac{\partial \ln q_{\text{rot}}}{\partial T} \right)_V \\ &= NkT^2 \left(\frac{\partial \ln \alpha T}{\partial T} \right)_V = NkT. \end{aligned}$$

Rotational contribution

- This is the same as the equipartition value of RT , in molar units. *The contribution to the heat capacity contribution is therefore R .*
- As with translational energy levels, making the assumption that the energy level spacing is much less than kT results in the equipartition values for U and C_V .
- When the temperature is so low that only the ground rotational level is occupied, $q_{rot} = 1$ and is independent of temperature. It therefore follows that $(\partial \ln q_{rot} / \partial T) = 0$ and so the rotational contribution to U and C_V is zero.
- Physically, *what this means is that the thermal energy, of the order of kT , is insufficient to promote any of the particles from the ground state. The system cannot therefore absorb energy, so its heat capacity is zero.*

Rotational contribution

- At intermediate temperatures the calculation is not so straightforward, as q_{rot} has to be computed term by term. The plot below shows how the rotational heat capacity varies with temperature:



- Note how the heat capacity quickly rises once the temperature approaches θ_{rot} , *levelling out at R when T exceeds θ_{rot} by a significant factor.*

Vibrational contribution

- For a diatomic the vibrational partition function (taking the $v = 0$ level as the energy zero) is

$$q'_{\text{vib}} = \frac{1}{1 - \exp\left(\frac{-\theta_{\text{vib}}}{T}\right)};$$

- note that *this was derived without making any assumption about the energy level spacing compared to kT* . The vibrational contribution to U is calculated in the usual way, although this time the algebra is more complicated:

$$\begin{aligned} U'_{\text{vib}} &= NkT^2 \left(\frac{\partial \ln q'_{\text{vib}}}{\partial T} \right)_V = NkT^2 \frac{1}{q'_{\text{vib}}} \left(\frac{\partial q'_{\text{vib}}}{\partial T} \right)_V \\ &= NkT^2 \left[1 - \exp\left(\frac{-\theta_{\text{vib}}}{T}\right) \right] \frac{d}{dT} \left[\left[1 - \exp\left(\frac{-\theta_{\text{vib}}}{T}\right) \right]^{-1} \right] \\ &= NkT^2 \left[1 - \exp\left(\frac{-\theta_{\text{vib}}}{T}\right) \right] (-1) \left[-\exp\left(\frac{-\theta_{\text{vib}}}{T}\right) \right] \frac{\theta_{\text{vib}}}{T^2} \left[1 - \exp\left(\frac{-\theta_{\text{vib}}}{T}\right) \right]^{-2} \\ &= \frac{Nk\theta_{\text{vib}} \exp\left(\frac{-\theta_{\text{vib}}}{T}\right)}{1 - \exp\left(\frac{-\theta_{\text{vib}}}{T}\right)} = \frac{Nk\theta_{\text{vib}}}{\exp\left(\frac{\theta_{\text{vib}}}{T}\right) - 1} \end{aligned}$$

- To go from the penultimate to the last line, the top and bottom of the fraction have been multiplied by **$\exp(\theta_{\text{vib}} / T)$** .

Vibrational contribution

- If we measured the energy from the bottom of the potential well there would be an additional term $U_0 = N \epsilon_0$ to include, where in this case $\epsilon_0 = 1/2 h\nu_0 = 1/2 k \theta_{vib}$. As was noted above, *U_0 is the energy which the system has at absolute zero due to the occupation of the ground states. This term is independent of temperature and so does not contribute to the heat capacity;* we will not consider it further.
- We now consider the limits of the equation derived above at low and high temperatures. Low temperature means $T \ll \theta_{vib}$; as $\theta_{vib} = h\nu_0 / k$ the inequality is equivalent to $kT \ll h\nu_0$ – in other words the vibrational energy level spacing is much greater than kT . In this limit θ_{vib} / T becomes very large, as does $\exp(\theta_{vib} / T)$. *The denominator becomes very large and so U'_{vib} goes to zero. The interpretation of this is that none of the excited vibrational energy levels are populated, so there is no vibrational contribution to the internal energy.*

Vibrational contribution

- *High temperature means $T \gg \theta_{\text{vib}}$, which implies that the vibrational energy level spacing is small compared to kT . Now θ_{vib} / T is small so the exponential can be approximated by the first two terms of the series expansion of e^x :*

$$U'_{\text{vib}, T \gg \theta_{\text{vib}}} = \frac{Nk\theta_{\text{vib}}}{1 + \frac{\theta_{\text{vib}}}{T} - 1} = NkT.$$

The series for e^x is $1 + x + x^2/2! + x^3/3! \dots$. If $x \ll 1$, a good approximation is to retain just the first two terms, so $e^x \approx 1 + x$.

- *The value in this limit is in accord with the prediction of the equipartition principle. Once again we see that the equipartition value is reached once the energy level spacing is small compared to kT .*

Vibrational contribution

- It is interesting to pursue this idea of low and high temperature limits further, this time starting from the vibrational partition function:

$$q'_{\text{vib}} = \frac{1}{1 - \exp\left(\frac{-\theta_{\text{vib}}}{T}\right)}.$$

- *If $T \ll \theta_{\text{vib}}$, θ_{vib} / T becomes very large, and so $\exp(-\theta_{\text{vib}} / T)$ becomes very small. The denominator in the expression for q' thus tends to 1 and so $q' = 1$. This is exactly as expected, since at low temperatures when kT is much smaller than the energy level spacing only the ground state term contributes to q' and hence $q' = 1$.*

Vibrational contribution

- If $T \gg \theta_{\text{vib}}$, θ_{vib}/T becomes very small, and so $\exp(-\theta_{\text{vib}}/T)$ can be well approximated as the first two terms in the series expansion of e^x :

$$q'_{\text{vib}} = \frac{1}{1 - \exp\left(\frac{-\theta_{\text{vib}}}{T}\right)} \approx \frac{1}{1 - \left(1 - \frac{\theta_{\text{vib}}}{T}\right)} = \frac{T}{\theta_{\text{vib}}}.$$

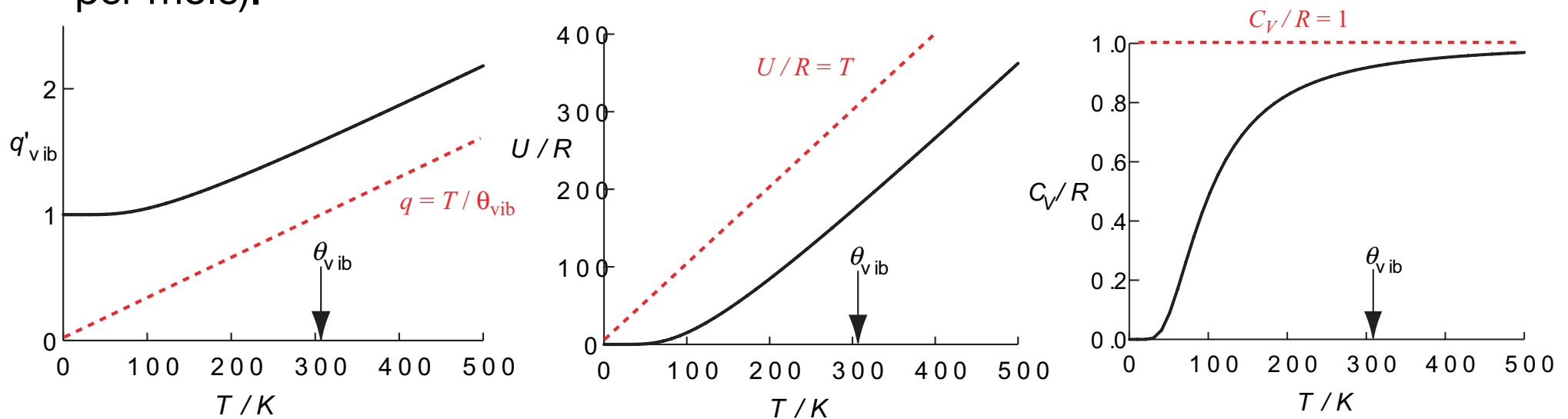
- It is easy to show that as $q' = aT$, the internal energy is NkT . This is the same result as we obtained by looking at the general expression for U_{vib}
- Starting from the expression of U_{vib} , we can, simply by differentiation, obtain a general expression for the vibrational contribution to the heat capacity.

$$\begin{aligned} C_{V,\text{vib}} &= \left(\frac{\partial U_{\text{vib}}}{\partial T} \right)_V = \frac{d}{dT} \left[Nk\theta_{\text{vib}} \left[\exp\left(\frac{\theta_{\text{vib}}}{T}\right) - 1 \right]^{-1} \right] \\ &= \frac{Nk\theta_{\text{vib}}^2}{T^2} \frac{\exp\left(\frac{\theta_{\text{vib}}}{T}\right)}{\left[\exp\left(\frac{\theta_{\text{vib}}}{T}\right) - 1 \right]^2}. \end{aligned}$$

- Using a similar approach to that used to find the limits of U_{vib} *it can be shown that at low temperatures ($T \ll \theta_{\text{vib}}$) C_V is 0 and that at high temperatures ($T \gg \theta_{\text{vib}}$) C_V is Nk ($= R$ per mole), as expected.*

Vibrational contribution

- The graphs below show the variation in q_{vib} , U_{vib} and $C_{V,vib}$ as a function of temperature for I_2 , which has $\theta_{vib} = 306$ K (the thermodynamic functions are per mole).



- At low temperatures we see that $q'_{vib} = 1$; as the temperature approaches θ_{vib} , q' starts to increase and quite quickly becomes linear in T , as predicted above. The internal energy starts at zero and only becomes significant as T approaches θ_{vib} ; the linear rise of U with T , predicted above, is quickly established. The heat capacity starts at zero and rises asymptotically to R ; its value is close to R by the time the temperature is θ_{vib} .
- Note that the linear rise in q' with T corresponds to the linear rise in U with T and the (asymptotic) flat part of the C_V curve.

Electronic contribution

- It is usually the case that only the ground electronic state is occupied and so the electronic contribution to the internal energy is simply $N \epsilon^0_{elec}$, where ϵ^0_{elec} *is the energy of the electronic ground state measured relative to some origin. This term is a significant contributor to the internal energy* (it is surely important in determining the values of equilibrium constants), *but as it does not depend on temperature it does not contribute to the heat capacity.*

Summary

- What we have found is that *a particular set of energy levels will give the equipartition value of the internal energy provided their spacings are much less than kT . If the spacings are much larger than kT , they make no contribution to the internal energy. When the spacings are comparable to kT , there is a transitional region when the internal energy begins to rise with temperature, but not linearly.*
- *When the energy level spacings are small compared to kT , we found that all the partition functions (for translation, rotation and vibration) are of the form $q = \alpha T^\beta$, where α is a temperature independent constant and β is a number. With this partition function the energy turns out to be $\beta \times NkT$ and the heat capacity $\beta \times Nk$.*

Course outline

- Entropy

Entropy

- In ‘classical’ thermodynamics entropies are determined from measurements of heat capacities as a function of temperature; it is necessary to make such measurements down to as close to absolute zero as possible. This is not very easy to do, so the calculation of entropies using spectroscopic data and the methods of statistical thermodynamics is an attractive option. For gases, such calculations can often give more precise estimates of the entropy than are available from heat capacity measurements. There is an important contrast here between classical thermodynamics, which tells us how to compute entropies from heat capacities, and statistical thermodynamics which allows us to compute entropies *ab initio*.

Separating different contributions

- We saw before that it is possible to separate the entropy into contributions from different sets of energy levels:

$$S_{\text{trans}} = Nk \ln q_{\text{trans}} - Nk \ln N + Nk + NkT \left(\frac{\partial \ln q_{\text{trans}}}{\partial T} \right)_V ,$$

$$S_{\text{rot}} = Nk \ln q_{\text{rot}} + NkT \left(\frac{\partial \ln q_{\text{rot}}}{\partial T} \right)_V ;$$

there are similar expressions to **S_{rot}** for the vibrational and electronic contributions.

- It is sometimes useful to rewrite these expressions to include the internal energy, **U** ; for each contribution to the internal energy the expression for **U** is the same:

$$U_{\text{trans}} = NkT^2 \left(\frac{\partial \ln q_{\text{trans}}}{\partial T} \right)_V \quad U_{\text{rot}} = NkT^2 \left(\frac{\partial \ln q_{\text{rot}}}{\partial T} \right)_V \text{ etc.}$$

Separating different contributions

- It follows that

$$\frac{U_{\text{trans}}}{T} = NkT \left(\frac{\partial \ln q_{\text{trans}}}{\partial T} \right)_V$$

which can be inserted into the right-hand side of the first equation in the last slide

$$S_{\text{trans}} = Nk \ln q_{\text{trans}} - Nk \ln N + Nk + \frac{U_{\text{trans}}}{T}$$

and likewise in the second equation to give

$$S_{\text{rot}} = Nk \ln q_{\text{rot}} + \frac{U_{\text{rot}}}{T}.$$

- In the same way, the vibrational contribution is

$$S_{\text{vib}} = Nk \ln q_{\text{vib}} + \frac{U_{\text{vib}}}{T}.$$

Note that when using this equation it is important to be consistent in the choice of energy zero for computing q_{vib} and U_{vib} .

Translational entropy

- The translational partition function in three dimensions is given:

$$q_{\text{trans}} = \left(\frac{2\pi mkT}{h^2} \right)^{\frac{3}{2}} V$$

and we have seen that this leads to an internal energy of **$3/2 NkT$** . It is therefore convenient to obtain the translational entropy

$$\begin{aligned} S_{\text{trans}} &= Nk \ln \left\{ \left(\frac{2\pi mkT}{h^2} \right)^{\frac{3}{2}} V \right\} - Nk \ln N + Nk + \frac{\frac{3}{2}NkT}{T} \\ &= Nk \left[\ln \left\{ \left(\frac{2\pi mkT}{h^2} \right)^{\frac{3}{2}} V \right\} - \ln N + \frac{5}{2} \right]. \end{aligned}$$

- This is often called the Sackur–Tetrode equation. Note that the expression we have used for the partition function was derived under the assumption that the energy levels are closely spaced compared to **kT** . Under these circumstances we can simply write down the value for **U_{trans}** as **$3/2 NkT$** using the equipartition principle; it is not necessary to compute it from **q_{trans}** .

Translational entropy

- The molar entropy is simply found by setting $N = N_A$:

$$S_{\text{trans,m}} = R \left[\ln \left\{ \left(\frac{2\pi mkT}{h^2} \right)^{\frac{3}{2}} V_m \right\} - \ln N_A + \frac{5}{2} \right]$$

- In this expression the molar volume, V_m , can be calculated using the ideal gas equation, **$pV_m = RT$** .
- N.B. Mind the units when doing the calculations.

Rotational entropy

- If the temperature is high enough that the rotational energy levels are spaced closely compared to kT , it was shown before that the rotational partition function for a diatomic is given by

$$q_{\text{rot}} = \frac{T}{\sigma \theta_{\text{rot}}}.$$

- Under these conditions the equipartition principle applies and the internal energy, U_{rot} , is NkT . Hence, the rotational entropy can be written

$$\begin{aligned} S_{\text{rot}} &= Nk \ln q_{\text{rot}} + \frac{U_{\text{rot}}}{T} \\ &= Nk \ln \frac{T}{\sigma \theta_{\text{rot}}} + \frac{NkT}{T} = Nk \left(\ln \left\{ \frac{T}{\sigma \theta_{\text{rot}}} \right\} + 1 \right) \end{aligned}$$

The same expression can be used for linear polyatomic molecules.

- For non-linear molecules we need to remember that in the high-temperature limit the internal energy due to rotation is $3 \times \frac{1}{2} NkT = \frac{3}{2} NkT$ as there are three rotational degrees of freedom (in contrast to the two degrees of rotational freedom for a linear molecule). The expression for the partition function was given in previous lectures.

Vibrational entropy

- The partition function for a harmonic oscillator (taking the ground state to have zero energy) is given

$$q'_{\text{vib}} = \frac{1}{1 - \exp\left(\frac{-\theta_{\text{vib}}}{T}\right)}.$$

- The corresponding internal energy is,

$$U'_{\text{vib}} = \frac{Nk\theta_{\text{vib}}}{\exp\left(\frac{\theta_{\text{vib}}}{T}\right) - 1}.$$

- The vibrational entropy is best calculated by finding q'_{vib} and U'_{vib} separately and then substituting these into

$$S_{\text{vib}} = Nk \ln q'_{\text{vib}} + \frac{U'_{\text{vib}}}{T}$$

For a polyatomic molecule, each normal mode makes a separate contribution to the entropy.

- If, as is often the case, $\theta_{\text{vib}} \gg T$, $q'_{\text{vib}} = 1$ and $U'_{\text{vib}} = 0$; the vibrational contribution to the entropy is zero. This is what we expect for the case when all the particles are in the ground state.

Electronic entropy

- It is usually the case that only the electronic ground state is occupied and that, if the energy of this state is taken to be zero, the electronic partition function is *just g_0 , the degeneracy of the ground state. The electronic contribution to the internal energy, U'_{elec} , is zero* so the electronic contribution to the entropy is given by

$$S_{\text{elec}} = Nk \ln q'_{\text{elec}} + \frac{U'_{\text{elec}}}{T} = Nk \ln g_0.$$

- Note that we were careful to choose the same energy zero for computing q and U .